ORIGINAL ARTICLE

Factor VII coagulant activity, factor VII −670A/C and −402G/A polymorphisms, and risk of venous thromboembolism

A. R. FOLSOM,* M. CUSHMAN,† S. R. HECKBERT,‡ T. OHIRA,* L. RASMUSSEN-TORVIK* and M. Y. TSAI§

*Division of Epidemiology & Community Health, School of Public Health, University of Minnesota, Minneapolis, MN; †Department of Medicine, University of Vermont, Burlington, VT; ‡Department of Epidemiology, University of Washington, Seattle, WA; and §Department of Laboratory Medicine and Pathology, University of Minnesota, Minneapolis, MN, USA

To cite this article: Folsom AR, Cushman M, Heckbert SR, Ohira T, Rasmussen-Torvik L, Tsai MY. Factor VII coagulant activity, factor VII −670A/C and −402G/A polymorphisms, and risk of venous thromboembolism. *J Thromb Haemost* 2007; **5**: 1674–8.

Summary. *Background:* Most epidemiological studies have found no association between levels of factor (F) VII:C and venous thromboembolism (VTE). Our Longitudinal Investigation of Thromboembolism Etiology (LITE) had, in contrast, reported an independent, increased risk of VTE after 7.8 years of follow-up for those with high baseline levels of FVII:C. *Objective:* To confirm whether FVII:C is associated with VTE after 12.6 years of follow-up and to examine whether two *FVII* gene polymorphisms (−670A/C and −402G/A) are related to VTE occurrence. *Methods:* In 19 091 LITE participants with no prior history of VTE or cancer, we measured FVII:C at baseline and identified 404 new VTEs. We also performed a nested case–control study to relate the polymorphisms to VTE ($n = 490$ without exclusion for cancer or prior VTE). *Results:* FVII:C was not independently associated with VTE occurrence after extended follow-up. Multivariable-adjusted rate ratios for VTE were 1.00, 1.00, 0.94, 1.00, and 1.38 (P-trend = 0.48) for the < 25th, 25th–49th, 50th–74th, 75th–94th, and ≥95th percentiles of FVII:C, respectively. The −670C and −402A alleles were in high linkage disequilibrium, and both were associated with greater FVII:C levels. However, neither polymorphism was associated with VTE occurrence. *Conclusion:* After extended follow-up, LITE offers little evidence that a greater FVII level is a risk factor for VTE.

Keywords: blood coagulation factors, factor VII, prospective study, pulmonary embolus, venous thrombosis.

Correspondence: Aaron R. Folsom, Division of Epidemiology & Community Health, School of Public Health, University of Minnesota, Suite 300, 1300 South Second Street, Minneapolis, MN 55454-1015, USA.
Tel.: +1 612 626 8862; fax: +1 612 624 0315; e-mail: folsom@epi.umn.edu

Received 9 March 2007, accepted 8 May 2007

Introduction

Abnormalities in procoagulation and anticoagulation, such as elevated plasma factor (F) VIII level and activated protein C resistance, increase the risk of venous thrombosis and pulmonary embolism [collectively termed venous thromboembolism (VTE)] [1]. Elevated FVII has not been associated with VTE risk in most prior studies [2,3]. The exception is our previous publication from the Longitudinal Investigation of Thromboembolism Etiology (LITE), which after a median follow-up of 7.8 years reported a rate ratio for future VTE of 2.4 (95% confidence interval 1.2–4.8) for a FVII coagulant activity (FVII:C) level above the 95th percentile compared with the lowest FVII:C quartile [4].

Variations in the level or activity of FVII have been linked to polymorphisms in the *FVII* gene. A −670A → C polymorphism is in tight linkage disequilibrium with a −402G → A polymorphism, and carriers have higher levels of FVII:C than non-carriers do [5–7]. Gene expression studies suggested that −670C, rather than −402A, is the functional variant responsible for raising levels of FVII:C [7]. Some studies suggest that the −670C and −402A alleles increase the risk of coronary heart disease [7–9]. To our knowledge, no study has examined the risk of VTE in relation to these *FVII* gene polymorphisms.

The purposes of this investigation were to determine (i) whether the short-term association observed in LITE between FVII:C and risk of VTE [4] persisted with extended follow-up and with more than twice as many VTE events, and (ii) whether VTE risk was increased in those with the −670C or −402A polymorphisms.

Methods

The LITE study is a prospective study of VTE occurrence in two pooled, multicenter, longitudinal, population-based cohort studies: the Atherosclerosis Risk in Communities (ARIC) study and the Cardiovascular Health Study (CHS). The LITE study design, methods, and VTE incidence rates have been described [10]. In brief, 15,792 ARIC participants aged 45–64 years at baseline in 1987–1989 and 5,201 CHS participants aged ≥65 years at baseline in 1989–1990 underwent assessments of cardiovascular risk factors. An additional 687 African Americans were recruited to CHS in 1992–1993, bringing the total LITE cohort size to 21 688. All participants provided informed consent and these studies were approved by the institutional review boards. At baseline, blood was drawn from fasting participants in the morning in both studies, promptly centrifuged at 30 000 *g*, and plasma and DNA were stored in freezers at −70 °C. Baseline thrombosis risk factors, including FVII:C and FVIII:C, body mass index (BMI; kg/m^2) and diabetes mellitus, were measured comparably in ARIC and CHS [11–13]. In ARIC, FVII:C and FVIII:C were measured at the University of Texas using FVII- or FVIII-deficient plasma and control materials obtained from George King Biomedical Inc. (Overland Park, KS, USA). Reliability coefficients (between-person variance divided by total variance) obtained from repeated testing of individuals over several weeks were 0.78 for FVII:C and 0.86 for FVIII:C. CHS used FVII-deficient plasma from Baxter-Dade (Bedford, MA, USA), FVIII-deficient plasma from Organon Teknika (Boxtel, the Netherlands), and control materials from George King Biomedical Inc. Between-batch reliability coefficients for blinded split specimens were 0.91 for FVII:C and 0.73 for FVIII:C.

Potential new VTE events were identified from baseline through 2002. Hospital records were obtained and VTE was validated by two physician reviewers using discharge summaries and imaging reports [10]. For the analysis of FVII:C with VTE, we excluded participants who were of ethnic groups scarcely represented in some field centers (n = 103), and then participants who had a history of VTE (n = 630) or cancer (n = 1,711) at baseline, were taking warfarin (n = 185), or were missing FVII:C data (n = 707). After all exclusions, 18,618 at-risk participants were included. We calculated rate ratios of FVII:C with VTE using Cox proportional hazards models.

In addition, LITE employed a nested case–control design to study prospective associations between parameters measured in stored blood specimens and VTE. Two controls without VTE were selected at random from the ARIC and CHS cohorts being followed, and frequency-matched to the VTE cases by age (5-year groupings), sex, race (African American or white), follow-up time, and study (ARIC or CHS). In total, 548 VTE cases and 1097 controls were selected. We measured FVII polymorphisms (−607A/C and −402G/A) [7], the G20210A polymorphism in the prothrombin gene [14], and FV Leiden (A1691G, R506Q) [15] using standard methods. D-dimer was measured using an immunoturbidometric assay on the Sta-R analyzer (Liatest D-DI, Diagnostica Stago, Parsippany, NJ, USA). The analytical coefficient of variation was 3.0%. For the nested case–control analysis, we excluded those who had not provided consent to use their DNA (n = 48), those who were taking warfarin at baseline (n = 20), those who were not white or African American (n = 6), and those who were missing FVII genotypes (n = 7). The associations of FVII polymorphisms with mean values of FVII:C, FVIII:C and D-dimer were assessed using ANOVA. Unconditional logistic regression was used to calculate odds ratios and 95% confidence intervals of VTE in relation to FVII polymorphisms.

Results

FVII:C level and VTE

In the full LITE cohort, followed-up for a median of 12.6 years, the age-, race-, and sex-adjusted rate of VTE increased across levels of FVII:C (Table 1). However, further adjustment for other risk factors for VTE in this cohort, i.e. BMI, diabetes mellitus and FVIII:C level, attenuated the rate ratios to non-significant levels, consistent with FVII:C being positively correlated with BMI and diabetes mellitus [16]. In fact, the positive and independent association between FVII:C and VTE was found in the earlier follow-up period through 1997, as reported previously [4]; however, absolutely no association was found between FVII:C and VTE in the follow-up period from 1997 to 2002, for which this rate ratio was 1.06 (95% CI 0.55–2.02) for above the 95th percentile compared with the lowest quartile of FVII:C.

There was also no association between FVII:C and VTE among most subsets of the sample, including men and women, obese and non-obese, and the subgroups in Table 2, other than

Table 1 Rate ratios (RR) and 95% confidence intervals (95% CI) of venous thromboembolism according to factor (F) VII:C levels (Longitudinal Investigation of Thromboembolism Etiology 1987–2002)

	FVII:C percentiles†					
	< 25th	25th–49th	50th–74th	75th–94th	≥ 95th	*P* for trend
FVII:C (range, %)	< 101	101–116	117–136	137–170	≥171	
Participants at risk (n)	4712	4505	4818	3651	932	
Cases (n)	91	93	104	84	32	
Person-years of follow-up	55 576	53 357	57 034	43 143	10 642	
Rate per 1000 person-years	1.6	1.7	1.8	1.9	3.0	
Age-, race-, and sex-adjusted RR (95% CI)	1.0	1.07 (0.80–1.43)	1.12 (0.84–1.49)	1.22 (0.90–1.66)	1.89 (1.25–2.87)	0.015
Multivariate-adjusted* RR (95% CI)	1.0	1.00 (0.74–1.36)	0.94 (0.69–1.27)	1.00 (0.72–1.37)	1.38 (0.90–2.12)	0.48

*Adjusted for age, race, sex, body mass index, diabetes mellitus, and FVIII:C. †Percentiles based on the combined Atherosclerosis Risk in Communities Study and Cardiovascular Health Study distributions.

Table 2 Multivariate-adjusted* rate ratios (RR) and 95% confidence intervals (95% CI) of venous thromboembolism (VTE) according to factor (F) VII:C levels (Longitudinal Investigation of Thromboembolism Etiology 1987–2002)

	FVII:C percentiles†						
	Events (*n*)	< 25th	25th–49th	50th–74th	75th–94th	≥95th	*P* for trend
Rate ratio (95% CI)							
ARIC	260	1.0	0.99 (0.69–1.40)	0.79 (0.55–1.13)	0.96 (0.66–1.41)	1.51 (0.92–2.48)	0.61
CHS	144	1.0	1.08 (0.59–1.98)	1.33 (0.76–2.35)	1.08 (0.58–2.00)	1.12 (0.47–2.67)	0.74
White participants	285	1.0	0.97 (0.68–1.37)	0.91 (0.64–1.29)	0.96 (0.67–1.39)	1.06 (0.62–1.82)	0.95
African American participants	119	1.0	1.12 (0.60–2.07)	1.00 (0.54–1.85)	1.05 (0.55–2.03)	2.28 (1.09–4.75)	0.18
Idiopathic VTE	163	1.0	1.25 (0.75–2.08)	1.24 (0.75–2.03)	1.21 (0.71–2.04)	1.78 (0.92–3.42)	0.21
Secondary VTE	241	1.0	0.89 (0.61–1.30)	0.79 (0.54–1.16)	0.90 (0.60–1.35)	1.18 (0.66–2.10)	0.88

ARIC, Atherosclerosis Risk in Communities Study; CHS, Cardiovascular Health Study. *Adjusted for age, race, sex, body mass index, diabetes mellitus and FVIII:C. †For percentiles see Table 1.

Table 3 Multivariate-adjusted odds ratios (OR) and 95% confidence intervals (95% CI) of venous thromboembolism in relation jointly to factor (F) VII:C and FVIII:C (Longitudinal Investigation of Thromboembolism Etiology 1987–2002)

FVIII:C (%)†	FVII:C (%)‡	Cases *n*	Non-cases *n*	OR*	95% CI
< 156	< 171	129	8427	1.00	Reference
< 156	≥171	3	335	0.51	0.16–1.59
≥156	< 171	217	8306	1.47	1.19–1.81
≥156	≥171	29	551	2.85	1.89–4.28

*Adjusted for age, sex, race, body mass index and diabetes mellitus. †Top quartile is cut-off value for high FVIII:C. ‡Top 95th percentile is cut-off value for high FVII:C.

Table 5 Frequencies of venous thromboembolism cases and controls in relation to factor (F) VII gene variants (Longitudinal Investigation of Thromboembolism Etiology 1987–2002)

	−670 A/C			−402G/A		
	AA *n* (%)	AC *n* (%)	CC *n* (%)	GG *n* (%)	GA *n* (%)	AA *n* (%)
White participants						
Cases	216 (60)	123 (34)	21 (6)	212 (59)	127 (35)	21 (6)
Controls	469 (63)	244 (33)	29 (4)	469 (63)	242 (33)	31 (4)
African American participants						
Cases	99 (76)	31 (24)	0 (0)	90 (69)	38 (29)	2 (2)
Controls	213 (80)	46 (17)	8 (3)	197 (74)	58 (22)	12 (4)

for African American participants, who had a 2-fold increased risk of VTE for FVII:C above the 95th percentile compared with the lowest quartile. In addition, elevated FVII:C and high FVIII:C (highest quartile) were synergistic (multiplicative interaction $P < 0.05$), such that those with both had a 2.9-fold increased risk of VTE compared with those who had neither factor elevated (Table 3).

FVII genotypes and VTE

In the nested case–control sample, the FVII −670A/C and −402G/A polymorphisms were in Hardy–Weinberg equilibrium among African American and white controls. They also were in very high linkage disequilibrium ($r^2 = 0.97$ in white controls and 0.72 in African American controls). As shown among controls in Table 4, the −670C and −402A alleles were associated with the highest mean baseline FVII:C level. These FVII alleles were not associated with D-dimer or FVIII:C in a dose-dependent manner. The FVII alleles were also not associated with other VTE risk factors (age, sex, BMI, diabetes mellitus, FV Leiden, or prothrombin G20210A) in this cohort.

In both white participants and African American participants, the −670C and −402A alleles were somewhat more common in cases than in controls (Table 5). Correspondingly, odds ratios relating these alleles to VTE were modestly greater than 1.0, particularly for white participants in relation to idiopathic events (Table 6); however, none of the odds ratios was statistically significant. Additional adjustment for FVII:C had little attenuating effect on the odds ratios, suggesting that any association between genotype and VTE risk is not explained by FVII:C level.

Table 4 Baseline risk factor levels according to factor (F) VII gene variants, Longitudinal Investigation of Thromboembolism Etiology controls

		−670 A/C				−402G/A			
	n†	AA	AC	CC	*P*-value‡	GG	GA	AA	*P*-value‡
Mean* ± SE									
FVII:C (%)	992	116 ± 1	129 ± 2	143 ± 4	< 0.0001	116 ± 1	129 ± 2	140 ± 4	< 0.0001
FVIII:C (%)	950	127 ± 1	133 ± 2	122 ± 6	0.04	127 ± 2	133 ± 2	125 ± 6	0.05
D-dimer (μg/mL)	903	0.54 ± 0.02	0.51 ± 0.03	0.71 ± 0.09	0.09	0.54 ± 0.02	0.52 ± 0.03	0.66 ± 0.08	0.30

*Adjusted for age, race, sex and parent study. †See Table 5 for genotype frequencies. ‡*P*-value for overall difference in means among genotypes.

Table 6 Odds ratios (OR) and 95% confidence intervals (95% CI) of venous thromboembolism* in relation to factor (F) VII gene variants (Longitudinal Investigation of Thromboembolism Etiology 1987–2002)

Group	−670 A/C AA	AC	CC	−402G/A GG	GA	AA
All (490 cases, 1009 controls)						
OR*	1.00	1.16	1.24	1.00	1.22	1.18
95% CI	Ref.	0.91–1.47	0.71–2.15	Ref.	0.96–1.54	0.70–2.00
All (490 cases, 1009 controls)						
OR†	1.00	1.09	1.19	1.00	1.15	1.14
95% CI	Ref.	0.85–1.39	0.68–2.11	Ref.	0.90–1.46	0.67–1.96
White participants (360 cases, 742 controls)						
OR‡	1.00	1.10	1.58	1.00	1.16	1.50
95% CI	Ref.	0.83–1.44	0.88–2.83	Ref.	0.89–1.52	0.84–2.67
White participants (360 cases, 742 controls)						
OR§	1.00	1.07	1.53	1.00	1.14	1.47
95% CI	Ref.	0.81–1.42	0.84–2.79	Ref.	0.86–1.51	0.81–2.65
White participants, idiopathic (149 cases, 742 controls)						
OR‡	1.00	1.15	1.87	1.00	1.16	1.75
95% CI	Ref.	0.79–1.68	0.88–3.99	Ref.	0.80–1.69	0.83–3.70

*Adjusted for age, race, sex and parent study. †Adjusted for age, race, sex, parent study and FVII:C. ‡Adjusted for age, sex and parent study; §Adjusted for age, sex, parent study and FVII:C. Ref., reference.

Discussion

In contrast with our earlier study with only 159 VTE events and 7.8 years of follow-up [4], LITE now finds no association between FVII:C and VTE risk after 12.6 years of follow-up. Basically, there was no association between FVII:C and VTE in the last 5 years of follow-up in this cohort. Our data imply that FVII:C is related only to short-term VTE risk. A weakening association with time might happen, for example, if a single, baseline FVII:C level does not adequately classify people as to long-term level. Alternatively, our earlier finding could have, by chance, been spurious. Other studies have suggested that FVII:C is not a VTE risk factor [2,3].

Our finding that in African American participants higher FVII:C was associated with a 2-fold greater VTE risk than lower FVII:C warrants additional confirmation. African Americans and whites have similar FVII:C levels [16–18], the main correlates of higher FVII:C being obesity and greater plasma lipids [16,19]. Likewise, our finding of possible synergy between elevated FVII:C and the known VTE risk factor, elevated FVIII:C, warrants confirmation because this was found in a supplemental analysis.

We also found that, although associated with higher FVII:C, the −670C and −402A FVII alleles were not associated with VTE risk. This contrasts with recent evidence that these alleles may increase risk of coronary heart disease [7–9]. We know of no other population-based studies of these polymorphisms and VTE, but other FVII polymorphisms have generally not been associated with incidence of VTE [1]. Unfortunately, we did not assess the R353Q polymorphism, which has also been associated with FVII:C variation and risk of coronary heart disease [7–9]. However, Zee *et al*. recently reported no association of this polymorphism with VTE in the Physicians' Health Study cohort [20]. Thus, studies to date do not suggest that variation in the *FVII* gene affects VTE risk, further implying that FVII level is not a VTE risk factor.

In conclusion, we found no association of FVII:C or two FVII polymorphisms with VTE occurrence in the long-term follow-up of this population-based cohort.

Acknowledgements

This work was supported by NHLBI grant R01 HL59367 (LITE), and contracts N01-HC 55015, N01-HC-55016, N01-HC-55018, N01-HC-55019, N01-HC-55020, N01-HC-55021, N01-HC-55022 (ARIC), and N01-HC-85079 to N01-HC-85086, N01-HC-75150, and N01-HC-45133 (CHS).

Disclosure of Conflict of Interests

The authors state that they have no conflict of interest.

References

1 Lowe GD. Can haematological tests predict cardiovascular risk? The 2005 Kettle Lecture. *Br J Haematol* 2006; **133**: 232–50.
2 Koster T, Rosendaal FR, Reitsma PH, van der Velden PA, Briet E, Vandenbroucke JP. Factor VII and fibrinogen levels as risk factors for venous thrombosis. A case-control study of plasma levels and DNA polymorphisms–the Leiden Thrombophilia Study (LETS). *Thromb Haemost* 1994; **71**: 719–22.
3 Lowe G, Woodward M, Vessey M, Rumley A, Gough P, Daly E. Thrombotic variables and risk of idiopathic venous thromboembolism in women aged 45–64 years. Relationships to hormone replacement therapy. *Thromb Haemost* 2000; **83**: 530–5.
4 Tsai AW, Cushman M, Rosamond WD, Heckbert SR, Tracy R, Aleksic N, Folsom AR. Coagulation factors, inflammation markers, and venous thromboembolism: The Longitudinal Investigation of Thromboembolism Etiology (LITE). *Am J Med* 2002; **113**: 636–42.
5 van't Hooft FM, Silveira A, Tornvall P, Iliadou A, Ehrenborg E, Eriksson P, Hamsten A. Two common functional polymorphisms in the promoter region of the coagulation factor VII gene determining plasma factor VII activity and mass concentration. *Blood* 1999; **93**: 3432–41.
6 Lindman AS, Pedersen JI, Hjerkinn EM, Veierød MB, Kavlie A, Arnesen H, Seljeflot I. The influence of the -401 G/T and -402 G/A polymorphisms of the coagulation FVII promoter on plasma levels of FVII. *Thromb Res* 2005; **116**: 313–20.
7 Carew JA, Basso F, Miller GJ, Hawe E, Jackson AA, Humphries SE, Bauer KA. A functional haplotype in the 5′ flanking region of the factor VII gene is associated with an increased risk of coronary heart disease. *J Thromb Haemost* 2003; **1**: 2179–85.
8 Lindman AS. Association between the factor VII haplotype containing the -402A allele and myocardial infarction in a population of elderly men at high risk for coronary heart disease. *Thromb Haemost* 2005; **94**: 226–7 (Letter).
9 Bozzini C, Girelli D, Bernardi F, Ferraresi P, Olivieri O, Pinotti M, Martinelli N, Manzato F, Friso S, Villa G, Pizzolo F, Beltrame F, Corrocher R. Influence of polymorphisms in the factor VII gene promoter on activated factor VII levels and on the risk of myocardial infarction in advanced coronary atherosclerosis. *Thromb Haemost* 2004; **92**: 541–9.
10 Cushman M, Tsai AW, White RH, Heckbert SR, Rosamond WD, Enright P, Folsom AR. Deep vein thrombosis and pulmonary embolism in two cohorts: the Longitudinal Investigation of Thromboembolism Etiology. *Am J Med* 2004; **117**: 19–25.

11 Tsai AW, Cushman M, Rosamond WD, Heckbert SR, Polak JF, Folsom AR. Cardiovascular risk factors and venous thromboembolism incidence: the Longitudinal Investigation of Thromboembolism Etiology (LITE) Study. *Arch Intern Med* 2002; **162**: 1182–9.
12 Folsom AR, Wu KK, Rosamond WD, Sharrett AR, Chambless LE. Prospective study of hemostatic factors and incidence of coronary heart disease: the Atherosclerosis Risk in Communities (ARIC) Study. *Circulation* 1997; **96**: 1102–8.
13 Cushman M, Cornell ES, Howard PR, Bovill EG, Tracy RP. Laboratory methods and quality assurance in the Cardiovascular Health Study. *Clin Chem* 1995; **41**: 264–70.
14 Poort SR, Rosendaal FR, Reitsma PH, Bertina RM. A common genetic variation in the 3′-untranslated region of the prothrombin gene is associated with elevated plasma prothrombin levels and an increase in venous thrombosis. *Blood* 1996; **88**: 3698–703.
15 de Ronde H, Bertina R. Laboratory diagnosis of APC-resistance: a critical evaluation of the test and the development of diagnostic criteria. *Thromb Haemost* 1994; **72**: 880–6.
16 Folsom AR, Wu KK, Davis CE, Conlan MG, Sorlie PD, Szklo M, for the Atherosclerosis Risk in Communities (ARIC) Study Investigators. Population correlates of plasma fibrinogen and factor VII, putative cardiovascular risk factors. *Atherosclerosis* 1991; **91**: 191–205.
17 Tracy RP, Bovill EG, Fried LP, Heiss G, Lee MH, Polak JF, Psat BM, Savage PJ, for the CHS Investigators. The distribution of coagulation factors VII and VIII and fibrinogen in adults over 65 years. Results from the Cardiovascular Health Study. *Ann Epidemiol* 1992; **2**: 509–19.
18 Green D, Ruth KJ, Folsom AR, Liu K. Hemostatic factors in the Coronary Artery Risk Development in Young Adults (CARDIA) Study. *Arterioscler Thromb* 1994; **14**: 686–93.
19 Cushman M, Yanez D, Psaty BM, Fried LP, Heiss G, Lee M, Polak JF, Savage PJ, Tracy RP, for the Cardiovascular Health Study Investigators. Association of fibrinogen and coagulation factors VII and VIII with cardiovascular risk factors in the elderly. *Am J Epidemiol* 1996; **143**: 665–76.
20 Zee RYL, Cook NR, Cheng S, Erlich HA, Lindpaintner K, Ridker PM. Polymorphism in the β_2-adrenergic receptor and lipoprotein lipase genes as risk determinants for idiopathic venous thromboembolism. A multilocus, population-based, prospective genetic analysis. *Circulation* 2006; **113**: 2193–200.